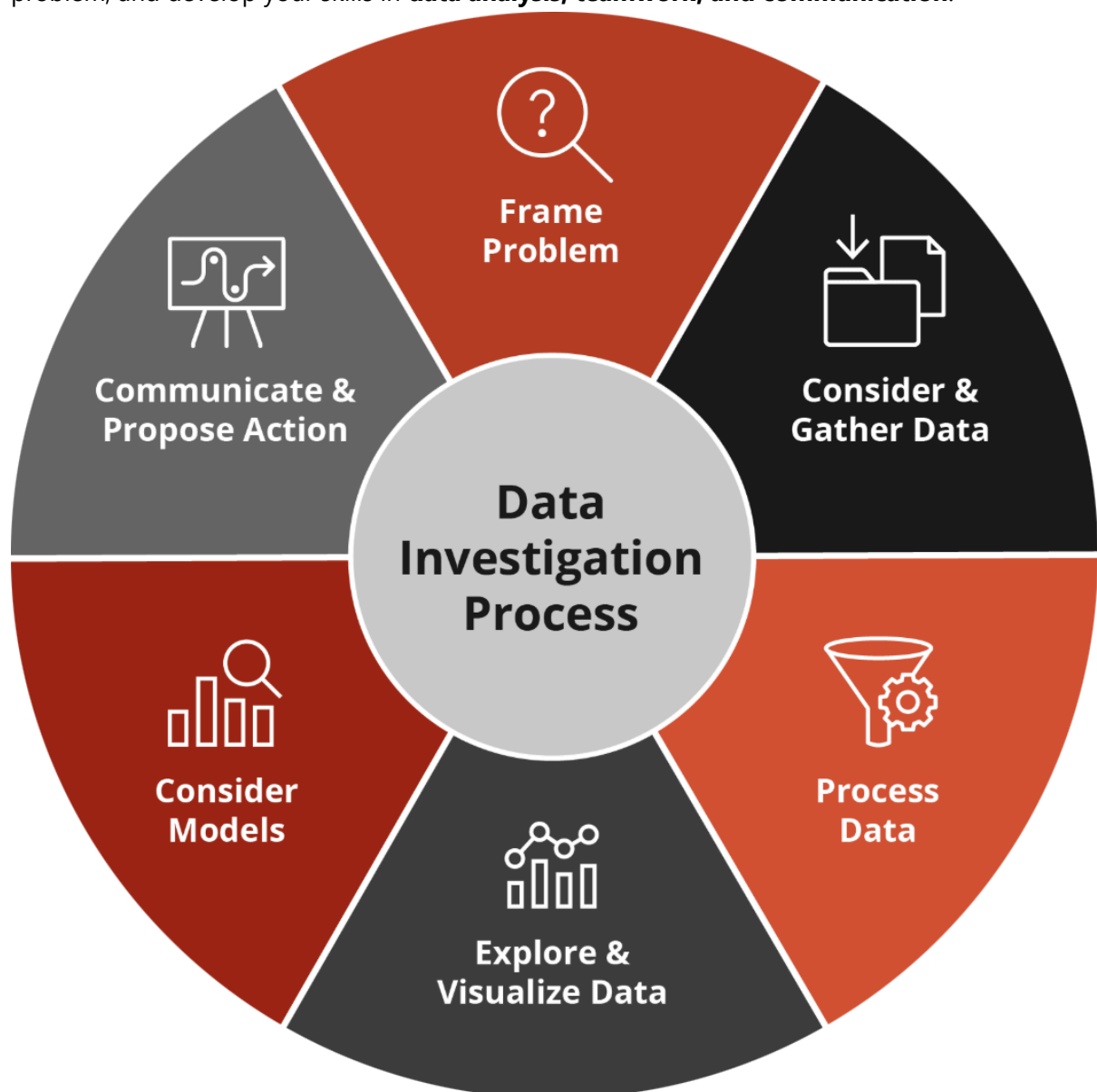


Group Project Info

Source: [Canvas — Group Project Info](#)

Welcome to the Group Project! In this assessment, you will work collaboratively to investigate a real-world dataset of your choice. Your team will define a meaningful research question, apply classification models to address it, and communicate your findings through both an oral presentation and written reports.

You will follow the **Data Investigation Process (DIP)** to complete a real-world classification problem, and develop your skills in **data analysis, teamwork, and communication**.



Project Milestones

You and your group members should complete the following tasks by the specified weeks.

Week 4: Group Registration

For the group project, students must form groups within their registered workshop. Groups will be created on Canvas. Your tutors will assist with the group registration process during the Week 4 workshop, so please make sure you attend your registered workshop.

- Groups will contain 5 students.
- Due to differences in workshop enrolment numbers, some workshops may have groups of 4* or 6 students.

Each group has a Group ID, which is made up of two parts:

- Workshop ID – indicates your workshop number W01 - W18
- Group Number – indicates your group within that workshop, G01 for example.

The project requirements and assessment will be applied uniformly to all groups, regardless of group size. For groups with 4 students, adjustments may be made to ensure the workload remains appropriate.

Ongoing: Group Communication and Collaboration

Group assignments mean that all members of the same group will initially receive the same mark, based on the quality of the group's submitted work as assessed against the published marking rubrics.

To ensure fairness and accountability in collaborative work, the final stage of the project involves a **non-graded Member Contribution Review**. If a student is identified as having contributed substantially less than their group members, their individual project mark may be moderated to better reflect their level of contribution.

To support effective teamwork and ensure transparent, traceable communication, all groups are expected to use the designated **Ed Chat platform (details to be updated later)**.

Week 7: Plan & Exploratory Data Analysis (EDA)

Objective

Successful modelling begins with understanding the data. Through project planning and EDA, you will investigate the characteristics of your dataset, identify potential challenges, and make informed modelling decisions.

Specific Tasks

1. Framing & Dataset Selection

- Identify a real-world classification task with clear local or global relevance.
- Formulate one specific research question.
- Motivation: individuals and/or groups will be affected by this problem and how your data investigation will help their decision-making processes.

2. Consider & Gather Data

- Document the data source(s) (e.g., the website links or journal papers).

- Dataset justification: Justify why chosen dataset addresses proposed research questions. Explicitly show how it meets **at least 3** of the given requirements (**Large, Messy, Complex, Integrated, Multi-class**).
 - *If Large Size (> 10,000 observations)*: Show row count and discuss computational considerations
 - *If Messy (missing values and/or potential outliers)*: Show missingness patterns with visualisation
 - *If Complex (> 100 features, mix of continuous and categorical variables)*: Show feature count and types, discuss dimensionality
 - *If Integration*: Document the data linkage process
 - *If Multi-class*: Show class distribution and discuss imbalance challenges
- Built-in R datasets, or datasets used in lectures or workshops are not permitted.
- Datasets found in Kaggle related to the following topics are not permitted, as they have been used extensively in previous cohorts: *credit cards, wine quality, airline satisfaction, rainfall, hospital admissions, road fatalities, used cars and Airbnb*.

3. Exploratory Data Analysis (EDA)

Conducting detailed exploratory data analyses supported by graphical and numerical summaries.

- *Univariate Analysis*: Distributions, missing values, outliers for key variables.
- *Bivariate Analysis*: Relationships between features and features; Relationships between features and outcome variables.
- *Data Visualisation*: Additional graphs/tables with insights

Note: The examples above are intended as a guide and are not exhaustive. You are encouraged to include any additional graphical or numerical analyses that help provide meaningful insights into the data and/or modelling decisions.

4. Project Planning

- Preprocessing Plan for handling your dataset's specific requirements to make it clean and ready for modelling.
- Model Plan: based on EDA, propose **5 classification models*** suited to the data/problem with justification.
- Evaluation Plan: propose and justify performance metrics for final model evaluation.
- Timeline with milestones and team responsibilities with back-up plans.

For groups of 4 students, the number of required classification models is reduced to 4.

5. References

- [Use APA 7th referencing style.](#)

6. Submission Details

- Submit one HTML document (maximum 6 pages) per group, uploaded by one group member on behalf of the group.
- Use code folding so markers can view the code when needed. *Submissions without any code will result in mark deductions!*

Report templates: [Quarto](#) ([Download Quarto](#)) and [RMarkdown](#) ([Download RMarkdown](#))

You can check the page limit by converting your HTML file to a PDF. Instructions can be found [here](#).

Week 12: Presentation

Objective

Communicate the outcomes of your data investigation to a non-specialist audience. This presentation assesses your ability to explain the problem being addressed, justify your modelling decisions, and present evidence-based insights and conclusions in a clear, engaging, and accessible manner.

Specific Tasks

Each group will deliver a live presentation during the Week 12 lecture or workshop, summarising the key findings from their analysis. Think of your presentation as telling a compelling data story: introduce the problem, explain how your team investigated it, highlight the most important findings, and demonstrate why your results matter. Your slides should cover the following topics (but feel free to include additional points where relevant).

- Project topic
- Research question and why it matters
- Your data and what it looks like
 - Include visualisations that highlight interesting patterns or stories in the data
- Data preprocessing
- Model refinement
 - How you improved your models (e.g. feature selection and parameter tuning)
- Model comparison
 - Key results supported by clear visuals
- Final insights
 - What you learned and how your findings link back to the research question

Presentation Logistics

- Use R Markdown or Quarto to build your slides (maximum 20 slides exclude the title page) and submit your slides [here](#).
- Presentation allocations TBA. You are expected to attend the full session.
- Each presentation is ~10 minutes (7 minutes live presentation + 3 minutes Q&A)

Presentation Tips

- Useful [documents for preparing your oral presentation](#) ([Download documents for preparing your oral presentation](#))
- Keep slides clean and simple — don't overload with text.
- Use graphs and tables that are:
 - Easy to read
 - Clearly labelled
 - Directly support your story

Focus on clarity, not technical detail. The audience should walk away understanding your project and why it matters.

Week 12: Report

Objective

Document your group's complete data investigation, demonstrating how you applied the Data Investigation Process (image above) to solve a real-world classification problem and communicate your findings effectively.

Specific Tasks

1. Introduction & Problem Statement/Motivation

2. Exploration Data Analysis

3. Model Implementation Details

- Preprocessing and splitting strategy
- Featurizing engineering
- Classification models fitted
- Hyperparameter tuning

4. Model Evaluation & Selection

- Summary of key findings models with performance metrics.
- Final model selection justification
- **Process reflection:** How did the DIP help structure your workflow? What would you do differently?

5. Conclusion & Insights

6. Reference

- [Use APA 7th referencing style](#)

Submission Requirements

- Submit one HTML document (maximum 10 pages)
- Use code folding so markers can view code if needed. *Submissions without any code will result in mark deductions!*

Week 13: Member Contribution Review

Objective

Ensure fairness and accountability in a collaborative group project.

The final stage of the group project is to complete the peer assessment and reflection tasks. It is optional in the sense that it does not contribute to your final grade.

The task requires you to rate yourself and your group members for all group assignments. This does not translate directly to a mark adjustment, but it will help us identify where mark adjustments may be required, particularly where there is a consistent signal coming from the group.

If your group had no issue, you can ignore this task.